

A Search-Adjusted Audit of Row-Sector Modulus Signals in Terminal Zeta-Zero Spacing Residuals

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Abstract

We re-examine a previously reported row-sector structure in terminal adjacent-zero spacing residuals of the Riemann zeta function. The earlier preprint identified an apparent base-6 row-sector rebound near a $\sim 10^3$ -row seam and reported modulus 1009 as the leading sector-amplitude ruler, with modulus 990 co-significant under local permutation guardrails. Subsequent validation showed that the principal companion paper's apparent h_1 phase-locking structure is better explained as a projection of terminal altitude localization. This motivated the corresponding question for the row-sector claim: does the reported modulus-1009 structure survive a look-elsewhere null that searches the same modulus family?

We answer this question negatively. A dense validation audit scans all moduli $M = 800, \dots, 1200$ and computes the sector-amplitude statistic $S_{\text{sector}}(M)$. Each permutation null shuffles tail labels and is allowed to scan the same 401 moduli, retaining its maximum sector amplitude. Under this search-adjusted null, neither modulus 1009 nor the best real modulus survives in any tested seam-scale configuration. In four primary fitted-ratio windows $W \in \{900, 1000, 1100, 1300\}$, modulus 1009 survives zero times, the best real modulus survives zero times, and modulus 1009 is never ranked in the top ten.

The replacement conclusion is therefore methodological rather than positive: the earlier row-sector signal is not a search-hardened discovery. It is consistent with a look-elsewhere artifact arising from sparse tail-sign populations searched over many moduli. Together with the revised companion result, this paper supports a unified interpretation: the surviving empirical object is terminal altitude architecture in highly compressed adjacent-spacing packets, while both continuous h_1 phase structure and discrete modulus-1009 sector structure were projection/search artifacts generated by exploratory coordinates.

Keywords: Riemann zeta zeros; zero spacing; row-sector residuals; modulus search; look-elsewhere effect; post-selection inference; terminal band; empirical audit.

1 Introduction

The present paper replaces an earlier positive interpretation of row-sector rebound dynamics in terminal zeta-zero spacing residuals. The original version reported a discrete row-index sector structure, amplitude-dominant under modulus 1009, co-significant under modulus 990, and empirically orthogonal to a previously reported h_1 packet organization. That interpretation is no longer maintained.

The change follows a blind-review objection concerning post-selection inference. The central issue is not whether a chosen modulus can look structured after being selected, but whether the structure survives a null model that is allowed to search the same modulus family. A local permutation null

conditional on a selected modulus does not address the look-elsewhere effect created by scanning many possible moduli, window sizes, metrics, and sectorizations.

This paper therefore asks a narrower and harder question:

Does the apparent modulus-1009 row-sector signal survive a dense search-adjusted null over $M = 800, \dots, 1200$?

The answer is no. Once the null is allowed to scan the same modulus range and keep its own best result, the real modulus-1009 signal is fully absorbed by the search-adjusted null envelope.

1.1 Relation to the revised companion paper

The companion paper originally framed rare extreme compressions as h_1 -phase-locked. Subsequent validation audits rejected that interpretation. The compressed packet population itself survived, but the h_1 corridor was explained as a projection of terminal altitude localization. The revised companion claim is therefore not “ h_1 phase-locking,” but rather:

highly compressed adjacent-spacing packets form a finite-horizon terminal altitude architecture.

This paper performs the corresponding cleanup for the row-sector continuation. Its conclusion is stronger as a negative result than as the earlier positive claim: the row-sector modulus signal does not survive search-adjusted modulus controls and should not be presented as a discovered sector engine.

1.2 Scope

All statements are empirical and finite-horizon. No asymptotic theorem is claimed. The analysis concerns the loaded terminal LMFDB-derived board spanning approximately 28.499547B to 30.610043B. The filename suffix 31p5B used in computational artifacts denotes a downloader ceiling, not a source-data endpoint.

2 Data and terminal board

2.1 Source board

All computations use the seam-stitched terminal checkpoint parquet,

`df_master_terminal_checkpoint_v16c_8000_31p5B.parquet`,

containing 935,111 terminal rows with altitude range approximately

$$28.499547\text{B} \leq T \leq 30.610043\text{B}.$$

The seam-stitching preserves adjacent-gap continuity across source shard boundaries. The suffix 31p5B is a retrieval ceiling only. A closure audit recorded SHA-256 checksum

`c80b841b1ebd0a602b45db9fa0544499fc6781858b70bbb1ec0940aeffe0fa76`.

2.2 Residual coordinates

The row-window residual pipeline uses spacing-ratio coordinates inherited from the companion audit chain:

$$\text{Gap_to_Pure_Ratio} = \frac{\Delta_k}{s_{\text{pure}}(T_k)}, \quad (1)$$

$$\text{Gap_to_Fitted_Ratio} = \frac{\Delta_k}{\hat{s}(T_k)}. \quad (2)$$

The primary dense-modulus validation uses Gap_to_Fitted_Ratio, which was the principal fitted-spine metric in the validation chain.

For a window of size W , row-window means are formed, a slow altitude trend is removed by a fitted log-altitude baseline, and residuals are robustly standardized. The tail population consists of rows satisfying

$$|z| \geq 2.575.$$

3 Original row-sector instrument

For modulus M , base-6 row sector is defined by

$$s_{6,M}(r) = \left\lfloor 6 \cdot \frac{r \bmod M}{M} \right\rfloor \in \{0, 1, 2, 3, 4, 5\}, \quad (3)$$

where r is the row-window midpoint.

For each sector s , let E_s be the number of expansion-tail events and C_s the number of compression-tail events. The sector imbalance vector is

$$I_s = \log \left(\frac{E_s + 1/2}{C_s + 1/2} \right), \quad (4)$$

and the sector-amplitude statistic is

$$S_{\text{sector}}(M) = \sqrt{\frac{1}{6} \sum_{s=0}^5 (I_s - \bar{I})^2}. \quad (5)$$

The earlier preprint used local permutation guardrails after candidate moduli had been identified. Those guardrails showed that several selected moduli, especially 1009 and 990, appeared nontrivial under local tests. The missing step was a search-adjusted null over the modulus-selection process itself.

4 Why the original positive claim is no longer maintained

The original row-sector claim had three components:

1. a sector-amplitude asymmetry exists in the terminal residual tail population;
2. modulus 1009 is the most consistent tested sector-amplitude ruler;
3. the row-sector structure is empirically orthogonal to the h_1 packet corridor.

The third component, orthogonality to h_1 , remains historically true in the narrow sense that the attempted h_1 -coupling tests did not confirm h_1 gating or local $h_1 \times \text{sector}$ modulation. However, the first two components no longer survive as positive claims. After the companion paper’s h_1 explanation was revised to altitude architecture, the row-sector signal required its own search-adjusted test. P86 supplies that test and rejects the positive modulus interpretation.

Thus the present paper is not a continuation of the old positive claim. It is its validation audit and replacement interpretation.

5 Dense modulus search-adjusted validation

5.1 P86 audit design

The P86 validation audit uses the following frozen design:

- primary metric: `Gap_to_Fitted_Ratio`;
- windows: $W \in \{900, 1000, 1100, 1300\}$;
- tail threshold: $|z| \geq 2.575$;
- modulus scan: $M = 800, \dots, 1200$;
- sectorization: base-6;
- permutations: 500 tail-label shuffles per configuration.

For each real configuration, $S_{\text{sector}}(M)$ is computed for all 401 moduli. For each permutation, tail signs are shuffled while row positions are held fixed, $S_{\text{sector}}(M)$ is recomputed for all 401 moduli, and the null retains

$$\max_{800 \leq M \leq 1200} S_{\text{sector}}^{\text{null}}(M).$$

The search-adjusted p -value compares the real selected statistic to this null distribution of maximums.

5.2 Primary dense-scan result

Table 1 gives the primary P86 summary. No tested configuration supports the earlier modulus-1009 claim.

W	Tail count	Best M	Best S	S_{1009}	Rank 1009	p_{best}	p_{1009}
900	112	1133	1.250916	0.476623	48	0.079840	1.000000
1000	104	1087	1.172635	0.385867	151	0.235529	1.000000
1100	93	1133	0.764446	0.395511	143	1.000000	1.000000
1300	71	1022	1.127564	0.591068	58	0.938124	1.000000

Table 1: P86 dense modulus search-adjusted validation. The null scans $M = 800, \dots, 1200$ for every permutation and keeps its maximum S_{sector} . Neither $M = 1009$ nor the best real modulus survives in any tested configuration.

Across these configurations:

$$\#\{W : p_{1009} \leq 0.05\} = 0,$$

$$\#\{W : p_{\text{best}} \leq 0.05\} = 0,$$

and

$$\#\{W : \text{rank}(1009) \leq 10\} = 0.$$

The P86 status is therefore

P86_NO_MODULUS_SURVIVES_LOOK_ELSEWHERE.

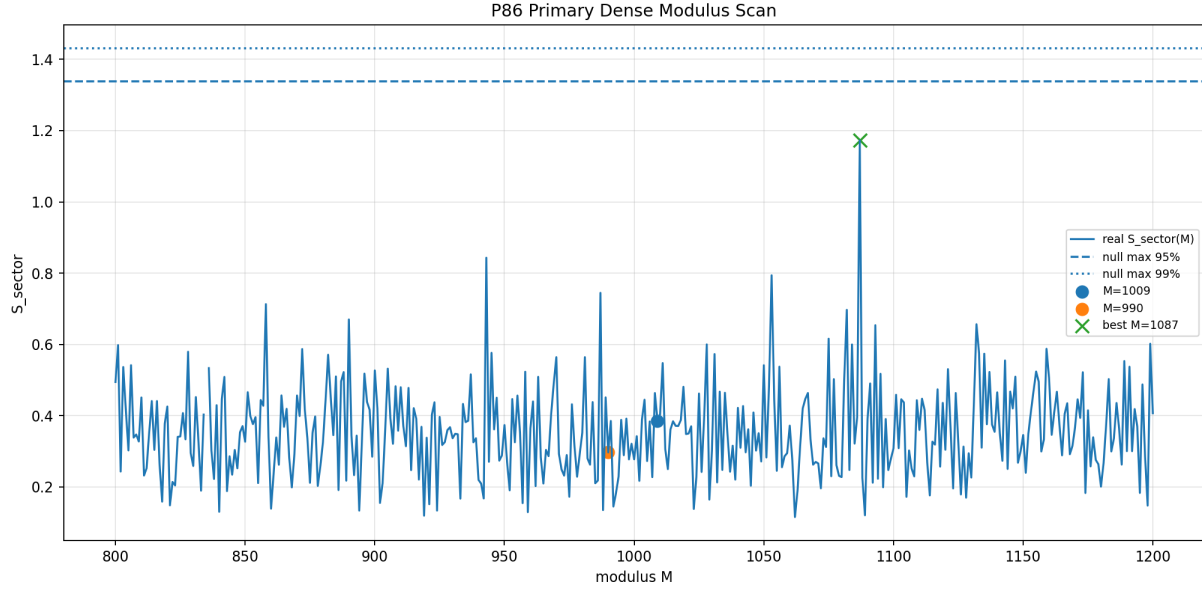


Figure 1: Primary dense modulus scan from P86. The real curve $S_{\text{sector}}(M)$ is plotted over $M = 800, \dots, 1200$, with markers for $M = 1009$, $M = 990$, and the best real modulus. The search-adjusted null ceiling is obtained from permutation replicates that scan the same modulus range.

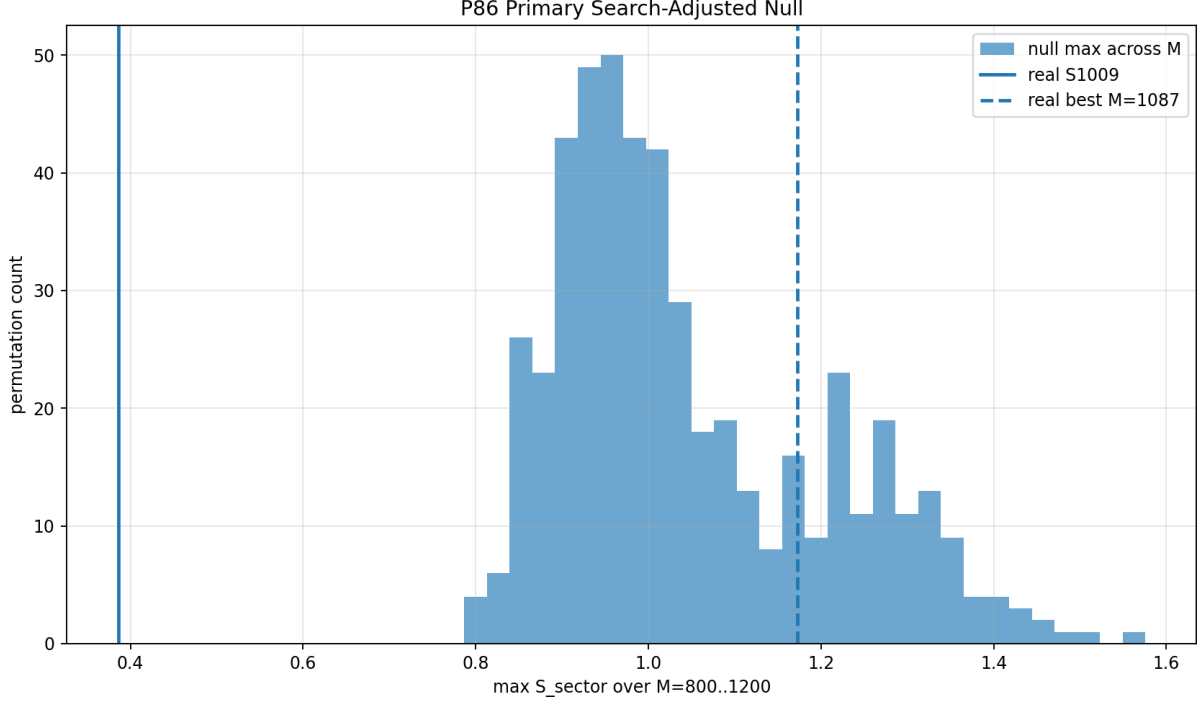


Figure 2: P86 search-adjusted null. Each null replicate shuffles tail signs and records the maximum sector amplitude over $M = 800, \dots, 1200$. The real modulus-1009 statistic lies well inside the null envelope.

5.3 Interpretation

The dense null shows that the earlier sector result is compatible with modulus selection over sparse tail populations. When hundreds of moduli are searched, it is expected that some modulus will produce a large apparent S_{sector} by chance. The correct null must search the same family, and under that null the signal disappears.

This is not a numerical bug or a failure of the sector statistic to compute what it was designed to compute. It is a post-selection failure: the local statistic was real conditional on a chosen modulus, but the chosen modulus was not protected against the look-elsewhere effect.

6 Claim transition ledger

Table 2 records the transition from the earlier row-sector preprint to the present replacement interpretation.

Original claim	Validation result	Replacement statement
A row-sector rebound structure exists as a positive discovery.	The apparent sector amplitudes do not survive dense search-adjusted modulus controls.	Rejected as a search-hardened positive claim.
Modulus 1009 is the leading sector-amplitude ruler.	In P86, $M = 1009$ survives in zero configurations and is never ranked in the top ten across the tested windows.	Modulus 1009 is not supported as a physical or distinguished modulus.
Modulus 990 is co-significant with 1009.	Under dense look-elsewhere correction, the modulus family as a whole fails to produce a surviving positive result.	The 990/1009 framing is retired.
The sector structure is orthogonal to h_1 .	The original h_1 claim itself is replaced by altitude architecture. The old orthogonality question no longer carries a positive sector result.	The episode becomes a methodological audit of projection artifacts.

Table 2: Transition from the earlier row-sector interpretation to the present search-adjusted interpretation.

7 Relation to terminal altitude architecture

The revised companion manuscript identifies a terminal altitude architecture in highly compressed adjacent-spacing packets. That architecture explains the old h_1 Arc80 corridor as an altitude projection. The present paper shows that the row-sector interpretation does not supply a second surviving mechanism.

The resulting unified picture is:

1. the terminal compression packet field is real and altitude-structured;
2. the continuous h_1 projection reveals that altitude structure but is not its mechanism;
3. the discrete row-sector modulus projection does not survive dense modulus look-elsewhere controls;
4. the surviving positive object is altitude architecture, not phase locking or modulus sectorization.

This reframing is important. It prevents the validation failure of h_1 and $M = 1009$ from being misread as failure of the packet object itself. The packet object survives; the projected coordinate stories do not.

8 Discussion

The negative result in this paper is scientifically useful because it closes a specific loophole. The earlier row-sector analysis used multiple local nulls, bin-resolution checks, and h_1 -interaction audits. Those tests were valuable, but they did not answer the decisive question: whether the selected modulus survives a null that searches the same modulus family.

P86 answers that question. It does not.

This result illustrates a general hazard in empirical number theory. Long arithmetic sequences are rich enough that many coordinate systems can be imposed on them. A modulus scan, a phase projection, or a geometric sectorization can appear structured if evaluated only after selection. The null must reproduce the selection process.

8.1 What remains from the original continuation paper

Several components remain useful:

- the formal row-sector statistic S_{sector} ;
- the interaction-residual framework for separating main effects from local h_1 -sector interactions;
- the bin-resolution calibration logic that prevented overclaiming local $h_1 \times \text{sector}$ structure;
- the documented failure modes of a plausible exploratory coordinate system.

These are methodological contributions, not positive evidence for a sector engine.

8.2 What does not remain

The following claims should no longer be made:

- modulus 1009 is a physical modulus;
- modulus 1009 is a search-hardened sector-amplitude leader;
- modulus 990 and 1009 define a validated scale-separated sector pair;
- the row-sector rebound is a confirmed positive structure independent of post-selection.

9 Limitations

The dense scan in P86 is intentionally focused on the primary fitted-ratio seam-scale family. It rejects the earlier positive claim under the relevant search-adjusted family $M = 800, \dots, 1200$, but it does not rule out every possible sectorization under every possible theoretical prior. A future theory that pre-specifies a modulus and a sector rule before seeing the data would require a different validation design.

The tail populations are sparse. This sparsity is part of why the look-elsewhere effect is severe: large apparent sector amplitudes can arise by chance when many moduli are scanned over a small number of tail events.

The result is finite-horizon. It concerns the loaded terminal LMFDB-derived range up to approximately 30.610B. It does not claim that all future modulus-based analyses must fail.

10 Conclusion

The earlier row-sector preprint reported an apparent modulus-1009 sector structure in terminal zeta-zero spacing residuals. A dense search-adjusted validation audit does not support that interpretation. When the null is allowed to scan the same $M = 800, \dots, 1200$ modulus family, neither modulus 1009 nor the best real modulus survives in any tested seam-scale configuration.

The row-sector result is therefore best understood as a look-elsewhere artifact produced by searching sparse tail-sign data over many moduli. In the unified replacement interpretation, the surviving empirical object is not a modulus sector engine but a terminal altitude architecture of highly compressed adjacent-spacing packets. The row-sector analysis remains valuable as a methodological audit: it shows how convincing a selected modulus can appear before search-adjusted validation is applied.

Data and Artifact Availability

The original row-sector preprint was deposited under DOI [10.5281/zenodo.20017507](https://doi.org/10.5281/zenodo.20017507). The principal companion preprint was deposited under DOI [10.5281/zenodo.19827222](https://doi.org/10.5281/zenodo.19827222). The replacement validation chain generated outputs in

`ForgeV16c_P86_DENSE_MODULUS_outputs`

including the dense real modulus curve, permutation null distribution, configuration summary, and flat manifest. The broader replacement scaffold is recorded in

`ForgeV16c_P87_PIVOT_SCAFFOLD_outputs.`

The source zero ordinates are attributed to LMFDB and Odlyzko zero tables.

Use of AI Tools

Generative AI tools were used for audit design review, code critique, adversarial claim evaluation, and draft assistance. All research design decisions, audit execution, interpretation, and final manuscript responsibility remain with Jeffery Huckstead.

Author Contributions

Jeffery Huckstead conceived and executed the audit chain, designed the claim ledger and validation protocol, and wrote the manuscript.

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